

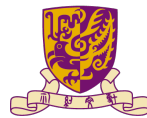
Network Programming

Introduction & Course Logistics

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SCHOOL OF
DATA SCIENCE
數據科學學院

About me

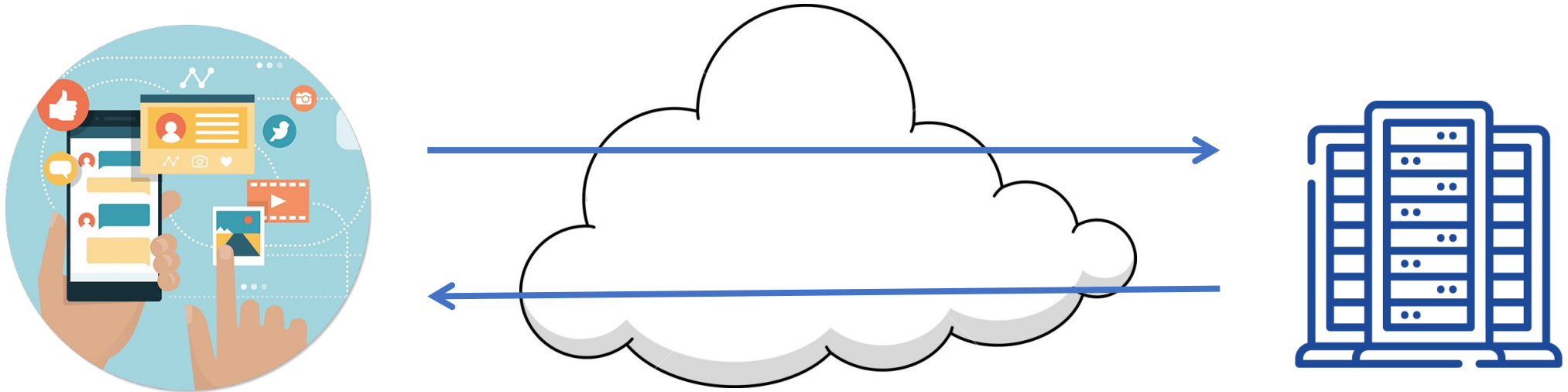
- Minchen Yu, Assistant Professor, SDS
 - Email: yuminchen@cuhk.edu.cn
 - Office: Zhixin 424
 - Office hour: Tuesday 3-4 PM
- Research interests
 - Cloud computing, distributed systems, AI infra and machine learning systems

Internet applications today

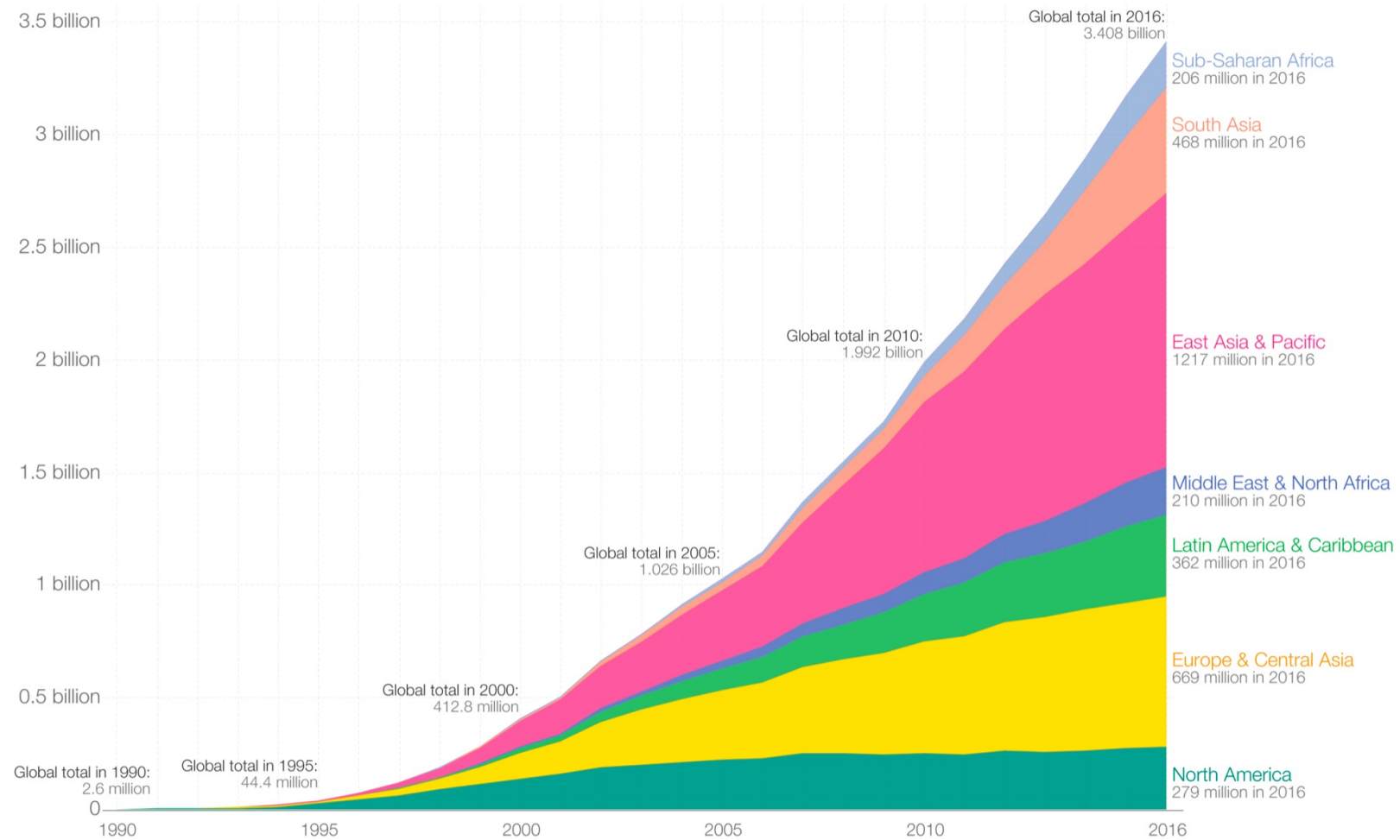
- Social media
- Online shopping
- Search engine
- Video platforms
- ...



Access services over Internet/network



Internet user growth



Data source: Based on data from the World Bank and data from the International Telecommunications Union. Internet users are people with access to the worldwide network. The interactive data visualization is available at [OurWorldinData.org](https://ourworldindata.org). There you find the raw data and more visualizations on this topic. Licensed under CC-BY-SA by the author Max Roser.

A view of service providers

- Establish network connections with users
- Handle many user requests
- Store and process a (large) volume of data
- ...



A view of service providers



Processes 20 PB a day (2008)
Crawls 20B web pages a day (2012)
Search index is 100+ PB (5/2014)
Bigtable serves 2+ EB, 600M QPS (5/2014)

300 PB data in Hive +
600 TB/day (4/2014)



S3: 2T objects, 1.1M
request/second (4/2013)

Datacenter



Copywrite: Google

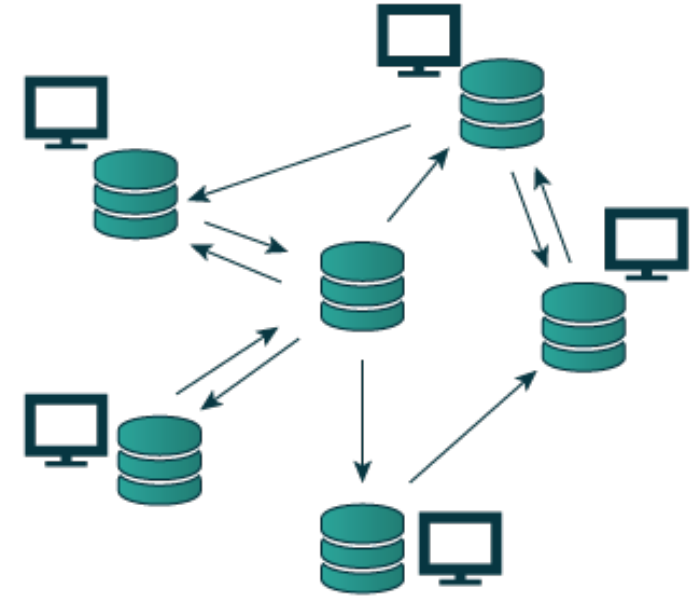
Datacenter

- >100k servers
- Costs in billions of dollars
- Geographically distributed



Distributed systems

- Store big data across multiple servers
- Communications between servers
- Recover from failures if happened
- ...



Example of distributed systems

- Distributed storage systems
 - Store a large amount of data in a distributed environment
- Communication frameworks
 - Collective communication primitives and implementation
- Distributed machine learning systems
 - Distributed large model training and serving

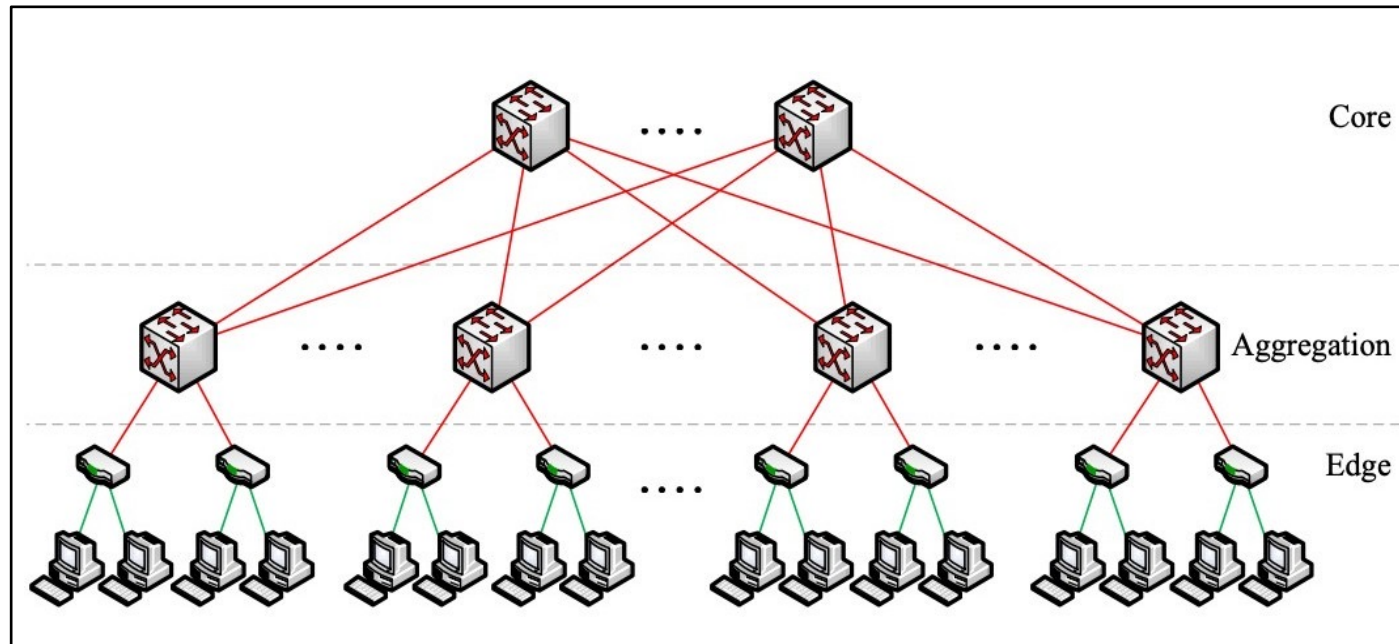


OPEN MPI

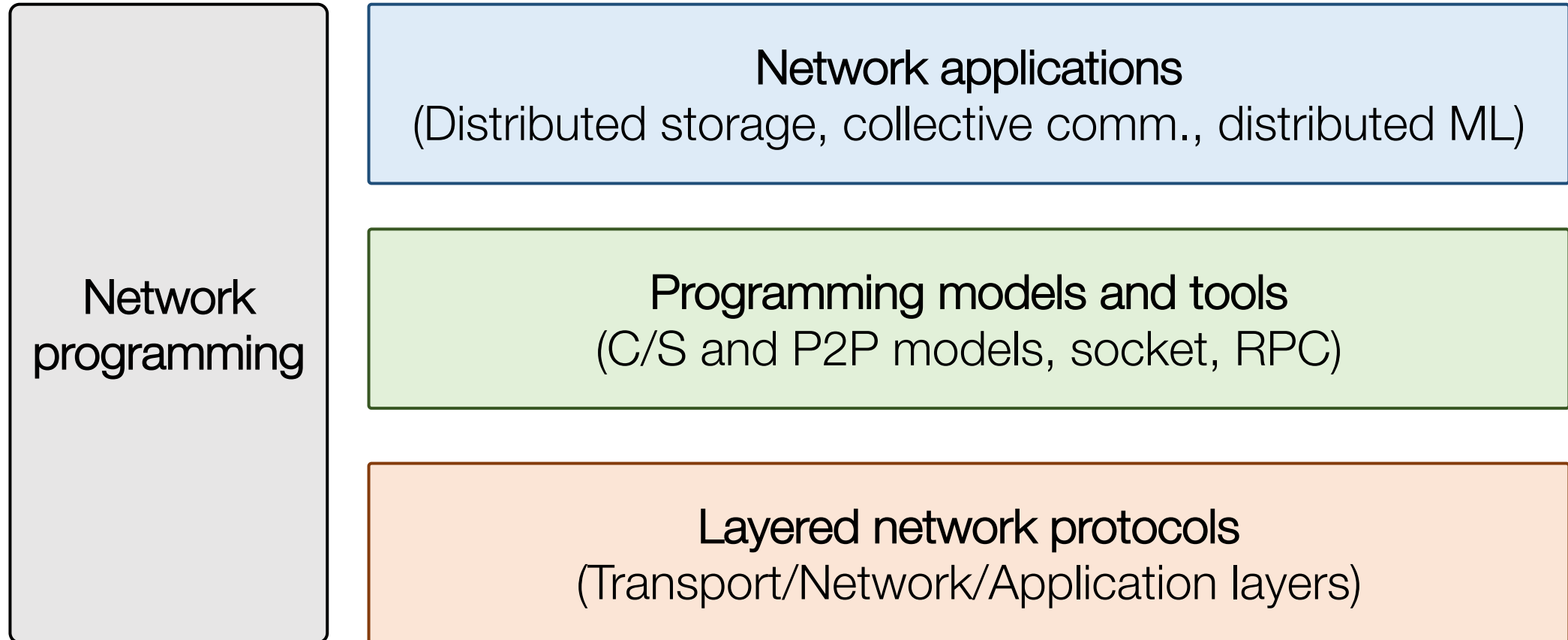


Data center networking

- Thousands of routers
- Hundreds of thousands of servers



Course overview



About the course

- Website: see Blackboard
- Announcements and course materials are posted online on a regular basis
- TAs
 - Yihao Zheng 225040496@link.cuhk.edu.cn
 - Yiyun Zheng 226040038@link.cuhk.edu.cn
 - Zhiyuan Tan zhiyuantan1@link.cuhk.edu.cn

Prerequisites

- Be comfortable with Unix/Linux
- Be comfortable with Python Programming
- Be comfortable with C/C++ Programming

Accept an email invitation to get
enrolled into an AWS Academy
Learner Lab

A note about EC2

- Amazon EC2 credit
 - \$50 USD credit from AWS Academy Classroom upon enrollment
 - AWS free tier: 750 hours per month for the **first 12 months**
- Shut down your virtual instances whenever you are done
- **Extra usage at your own cost.** We **cannot** financially help you in any means!

References

- References
 - Computer Networks: A Systems Approach
 - Unix Network Programming, Volume 1: The Sockets Networking API
 - Other online materials (check Blackboard)
- Learn things **online**

Assessment

- Programming assignments
 - 60%
- Final exam
 - 40%

Objectives

- Learn and understand the following aspects
 - Network layered design
 - Programming techniques for network communications
 - Distributed storage systems
 - Collective communication and distributed machine learning
 - Data center networking

Tips

- Attend the lecture and tutorial
- Get your hands dirty
- Learn things online
- Do all assignments by yourself
- Do well in the final

Learn programming by trying it yourself!

Academic honesty

- In short, **don't cheat!**
- **Don't** copy code or solutions from your classmates or third-party sources, and **don't** let others copy yours. Both cases are plagiarism and penalized in the same way

Protocol for Plagiarism

- TAs will detect possible plagiarism in your code/reports.
- Suspicious cases will be directly reported to the school.
- Minimum penalty: zero mark for the assignment.

Tentative lecture schedule

Week	Topic
1	Logistics, Computer Network Overview
2	Transport Layer I
3	Transport Layer II
4	Socket Programming
5	Network Layer
6	P2P
7	Distributed Storage System I
8	Application Layer
9	RPC
10	Distributed Storage System II
11	Collective Communication
12	Distributed Machine learning
13	Data Center Networking, Review
14	Final Exam

Subject to change.
Check back frequently!

Tentative tutorial schedule

- Thursday 6-7 PM
- Teaching C 414
- The first tutorial starts next week (9.17)
- Tutorial may be skipped with notification in advance

Any questions?